

Alison Yim

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EDUCATION

Lehigh University - P.C Rossin College of Engineering & College of Business

August 2020 - May 2024

Bachelor of Science, Computer Science and Business Honors Program

SKILLS

- **Languages:** C#, Python, Java, C++, C, SQL, JavaScript, TypeScript, PowerShell, KQL, Rust
- **Frameworks & Cloud:** Azure, ASP.NET Core, CosmosDB, PostgreSQL, React, MongoDB, Node.js, Google Cloud Platform
- **Tools & Libraries:** CI/CD (GitHub Actions, Azure DevOps), Docker, Git, Linux, Pandas, NumPy, TensorFlow, MATLAB

EXPERIENCE

Microsoft

Redmond, WA

Software Engineer

July 2024 - Present

- Engineer C# Azure control-plane infrastructure, reducing deployment overhead by ~80% via automated ARM-based workflows
- Develop distributed, concurrent backend services and scalable CosmosDB data models with production observability
- Migrate 50% of production service operations to event-driven RESTful APIs with unit and integration tests
- Extend backend validation pipelines with structured telemetry and request tracing to audit IAM/RBAC changes

Nano Human Interfaces Lab

Bethlehem, PA

Undergraduate Researcher

August 2023 - May 2024

- Built Python (pandas) pipeline analyzing 833 data points into 7 MATLAB visualizations guiding a Unity AR app
- Classified open-ended responses from 110 users into themes via text analysis, quantifying top barriers
- Synthesized 200+ research sources and 24 user-testing insights into evidence-based product content

Microsoft

Redmond, WA

Software Engineering Intern

May - August 2023

- Designed a self-service backend platform enabling partner teams to subscribe to event-driven CRUD notifications through APIs
- Built fault-tolerant event-driven pipeline processing asynchronous workloads, eliminating 99.7% of manual operations
- Enhanced Azure Monitor regional onboarding infrastructure, improving deployment confidence by 20%

Microsoft

Redmond, WA

Explore Intern (Product Management & Software Engineering)

May - August 2022

- Designed VM-placement algorithm optimizing valid test configurations that maximize memory, CPU, and bandwidth
- Automated VM deployment and validation via PowerShell and Kusto, cutting log analysis 99.84% (2 hours to 15 seconds)
- Led 3-intern team as PM with Agile deliverables and SWE/PM syncs; presented to Azure Server Performance

PROJECTS

Sip & Score (Matcha Log), [allyyim.github.io/matchaRatings/](https://github.com/allyyim/matchaRatings/)

May - September 2026

- Architected offline-first PWA with 30+ JWT-secured REST APIs and CI/CD, cutting cold-start latency by 30%
- Built real-time recommendation engine scoring 12 flavor and 5 body dimensions with weighted SQL filtering
- Technologies used: React, TypeScript, Bootstrap, TensorFlow.js, Node.js, GitHub Actions, PostgreSQL (Supabase), Render

ADO Prism (Microsoft MRTA Hackathon 2025), https://github.com/allyyim/MRTAthon_hack2025

December 2025

- Built an AI-powered internal engineering assistant to query pull requests and reduce manual knowledge triage by ~58%
- Designed an NLP pipeline that classified critical PR comments using semantic and keyword-based analysis
- Technologies used: HTML, Azure OpenAI, C#

Predictive Autoscaling (Microsoft Hackathon 2024) <https://github.com/allyyim/PredictiveAutoscaling>

September 2024

- Extended Azure Monitor predictive autoscaling by adding memory utilization and growing training data 43% (7 to 10 days)
- Tuned ML.NET regression model (L1/L2 regularization, normalization) to refine autoscaling predictions
- Technologies used: C#, ML.NET (TimeSeries), Azure Monitor, OxyPlot